



Introduction to Biostatistics

MSBD520

Fall 2026

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Housekeeping

- Office hours

Wednesdays 11 am - noon CT

Fridays 3 - 4 pm CT

- I have created a Class link for our weekly class at the top of the Blackboard course.

- Starting next week, any assignments will incur a penalty for late submissions.



Elementary Concepts On Statistics

Branches of Statistics - Descriptive and Inferential Statistics

Populations and Samples

Variables and Constants

Types of Study

Scales of Measurement

Probability, random variables, and their distributions

Describing Data - Shape

Frequency Distributions

Measures of Central Tendency

Measures of Dispersion or Variability

Data Standardization - Z-scores

Estimation, Confidence Interval, and Effect Size

How do we do the statistics?

What are the steps we have to take?

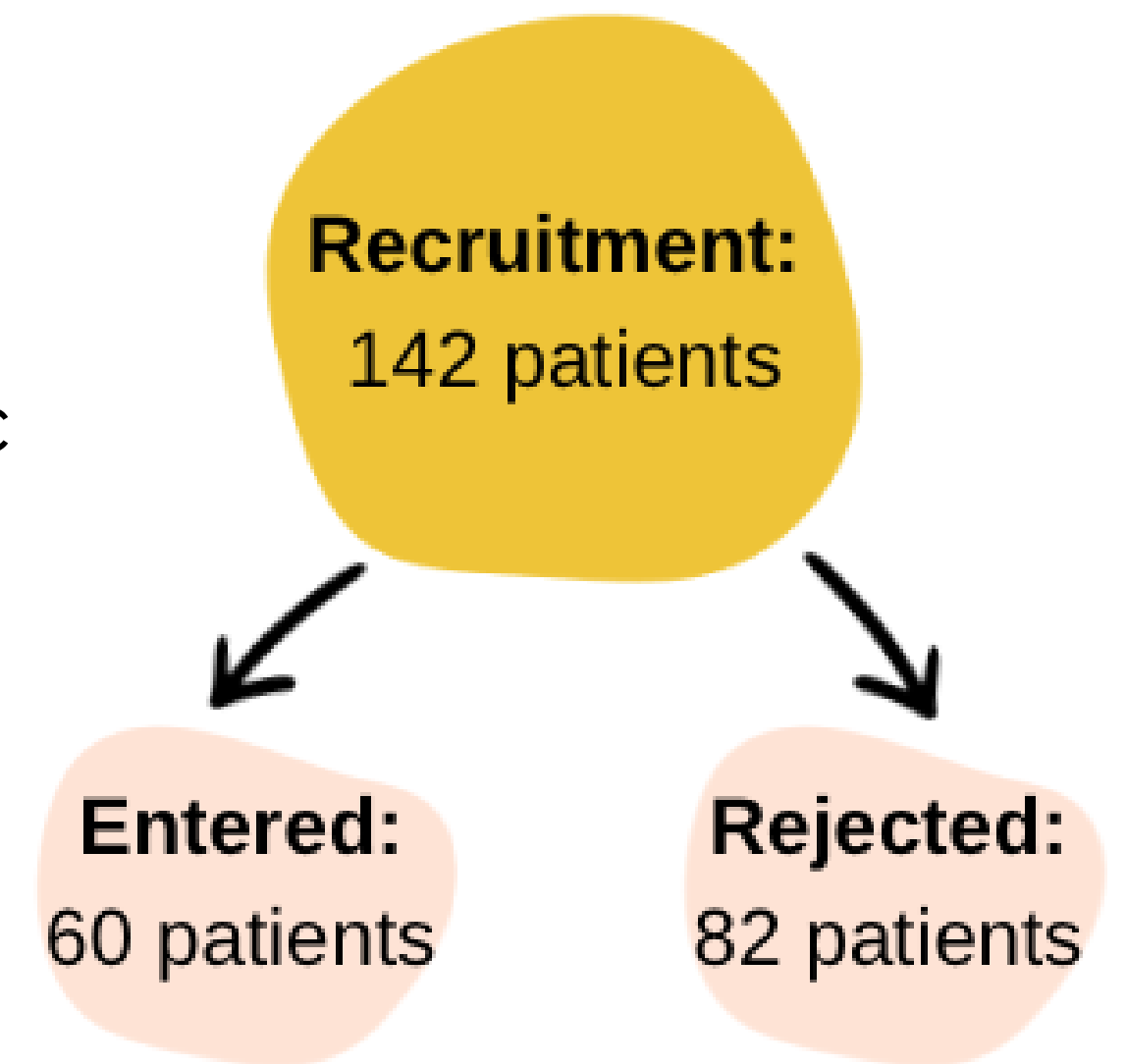
An example

Treating Chronic Fatigue Syndrome

Objective. Evaluate the effectiveness of cognitive-behavior therapy for chronic fatigue syndrome.

Participant pool. 142 patients who were recruited from referrals by primary care physicians and consultants to a hospital clinic specializing in chronic fatigue syndrome.

Actual participants. Only 60 of the 142 referred patients entered the study. Some were excluded because they didn't meet the diagnostic criteria, some had other health issues, and some refused to be a part of the study.

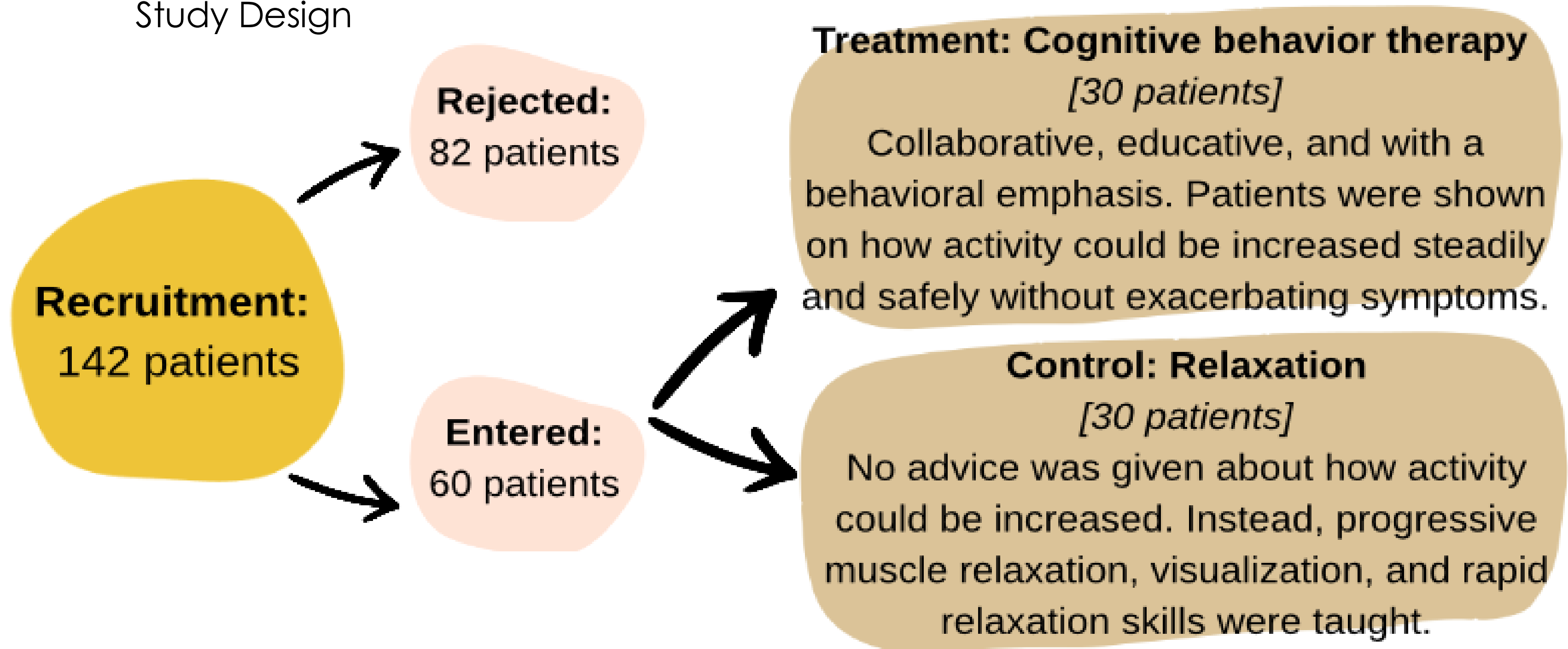


Deale, et. al. 1997. Cognitive behavior therapy for chronic fatigue syndrome: A randomized controlled trial. The American Journal of Psychiatry 154:3.

An example

Treating Chronic Fatigue Syndrome

Study Design



An example

Treating Chronic Fatigue Syndrome

The table below shows the distribution of patients with good outcomes at 6-month follow-up. Note that 7 patients dropped out of the study: 3 from the treatment group and 4 from the control group.

		<i>Good outcome</i>		Total	
		Yes	No		
<i>Group</i>	Treatment	<u>19</u>	8	<u>27</u>	- 3 missing
	Control	<u>5</u>	21	<u>26</u>	- 4 missing
	Total	24	29	53	

Proportion with good outcomes in treatment group

$$19/27 \approx 0.70 \rightarrow 70\%$$

Proportion with good outcomes in control group

$$5/26 \approx 0.19 \rightarrow 19\%$$

Do the data show a "real" difference between the groups?

Understanding the results

Suppose you flip a coin 100 times. What are the chances a coin land heads in any given coin flip?

50%. But we probably won't observe exactly 50 heads. This type of fluctuation is part of almost any type of data generating process.

In the previous study, the observed difference between the two groups ($70 - 19 = 51\%$) may be real or may be due to natural variation. Since the difference is quite large, it is more believable that the difference is real.

Understanding the results

Q: Are the results of this study generalizable to all patients with chronic fatigue syndrome?

These patients had specific characteristics and volunteered to participate in this study; therefore, they may not be representative of all patients with chronic fatigue syndrome. While we cannot immediately generalize the results to all patients, this first study is encouraging.

The method works for patients with a narrow set of characteristics, and that gives hope that it will work, at least to some degree, with other patients.

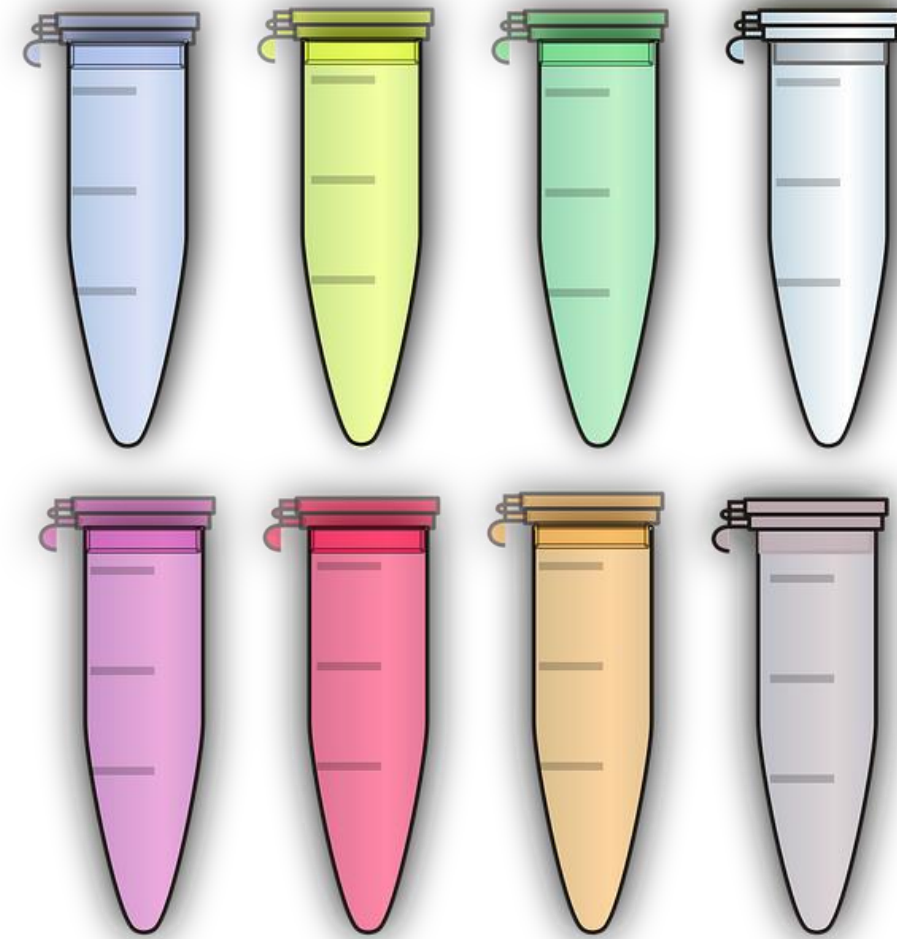
Topics in Intro to Data

Types of variables

Relationship between variables

Data Collection

Experiments





Data Basics

An Example: Classroom Survey

A survey was conducted on students in an introductory statistics course. Below are a few of the questions on the survey, and the corresponding variables the data from the responses were stored in:

- **gender**: What is your gender?
- **intro_extro**: Do you consider yourself an introvert or an extrovert?
- **sleep**: How many hours do you sleep at night, on average?
- **bedtime**: What time do you usually go to bed?
- **countries**: How many countries have you visited?
- **dread**: On a scale of 1-5, how much do you dread being here?

An Example: Classroom Survey

Data was collected on students a statistics class on a variety of variables:

Stu.	gender	intro_extra	...	dread
1	male	extravert	...	3
2	female	extravert	...	2
3	female	introvert	...	4
4	female	extravert	...	2
:	:	:	:	:
86	male	extravert	...	3

Data

Data can be organized in a data matrix

Variable:
Recorded characteristics

Data
Matrix

Stu.	gender	intro_extra	...	dread
1	male	extravert	...	3
2	female	extravert	...	2
3	female	introvert	...	4
4	female	extravert	...	2
:	:	:	:	:
86	male	extravert	...	3

Observation:

Types of Variables

Numerical

Continuous

(measured characteristics)

weight time
height temperature

Discrete

(can be counted)

age # of children
of residents

Categorical

Nominal (regular)

(defined categories)

gender, marital status
political party, job status,
eye color

Ordinal

(ordered categories)

rating $\begin{cases} \text{good, better, best} \\ 1-5 \\ \text{low, medium, high} \end{cases}$

Let's define our variables

	<i>categorical</i>	<i>numerical cont</i>	<i>categorical ordinal</i>	<i>numerical discrete</i>	<i>categorical ordinal / numerical</i>
	gender	sleep	bedtime	countries	dread
1	male	5	<u>12-2</u>	13	3
2	female	7	<u>10-12</u>	7	2
3	female	5.5	12-2	1	4
4	female	7	12-2		2
5	female	3	12-2	1	3
6	female	3	12-2	9	4

Note: Some of the variables can be categorical or numerical. It depends on how we set the data during collection.

Let's define our variables

What type of variable is a telephone area code?

- (a) numerical, continuous
- (b) numerical, discrete
- ☒ (c) categorical
- (d) categorical, ordinal

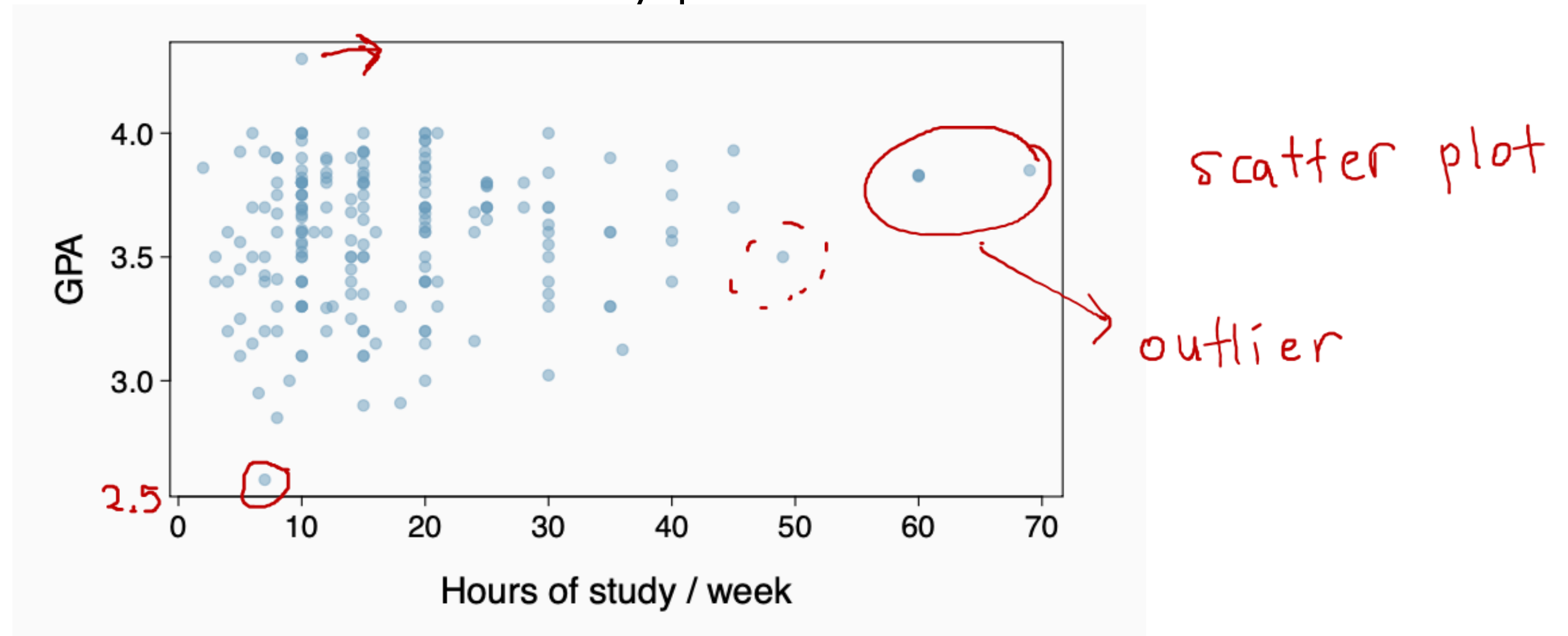
615
629
734



Because area code is not in order, and the number of the area code does not represent “large” or “small”, we cannot say this variable is a numerical variable.

Relationship between Variables

Does there appear to be a relationship between GPA and the number of hours that students study per week?



Can you spot anything unusual about any of the data points?
>> There is one student with $\text{GPA} > 4.0$, this is likely a data error.

Relationship between Variables

To identify the explanatory variable in a pair of variables, identify which of the two is suspected of affecting the other:

explanatory variable $\xrightarrow{\text{might affect}}$ response variable

Labeling variables as explanatory and response does not guarantee the relationship between the two is actually causal, even if there is an association identified between the two variables.

We use these labels only to keep track of which variable we suspect affects the other.

Relationship between Variables

Example: In the maternal investment study conducted on frogs, researchers collected measurements on egg clutches and female frogs at 11 study sites, located at differing altitudes, in order to investigate how egg size and clutch size vary with altitude.

Identify the response and explanatory variables in the study.

Response variable: egg size, clutch size

Explanatory variable: altitude

Not in this study
site location
frog gender
body size



Data Collection

Types of Data Collection

Observational Studies

Researchers collect data in a way that does not interfere with how the data arises.

||

Can provide evidence of a naturally occurring association between variables, but cannot themselves show a causal connection

Experiment

Researchers randomly assign subjects to various treatments.

||

Establish causal connection between explanatory and response variables

Types of Data Collection

Let's look at these example:

A. A study took random sample of adults and asked them about their bedtime habits. The data showed that people who drank a cup of tea before bedtime were more likely to go to sleep earlier than those who didn't drink tea.

observational study

B. A study randomly assigned volunteers to one of two groups:

- One group was directed to use social media sites as they usually do.
- One group was blocked from social media sites.

experiment

The researchers looked at which group tended to be happier.

A. Another study took a random sample of people and examined their social media habits. Each person was classified as either a light, moderate, or heavy social media user. The researchers looked at which groups tended to be happier.

observational study

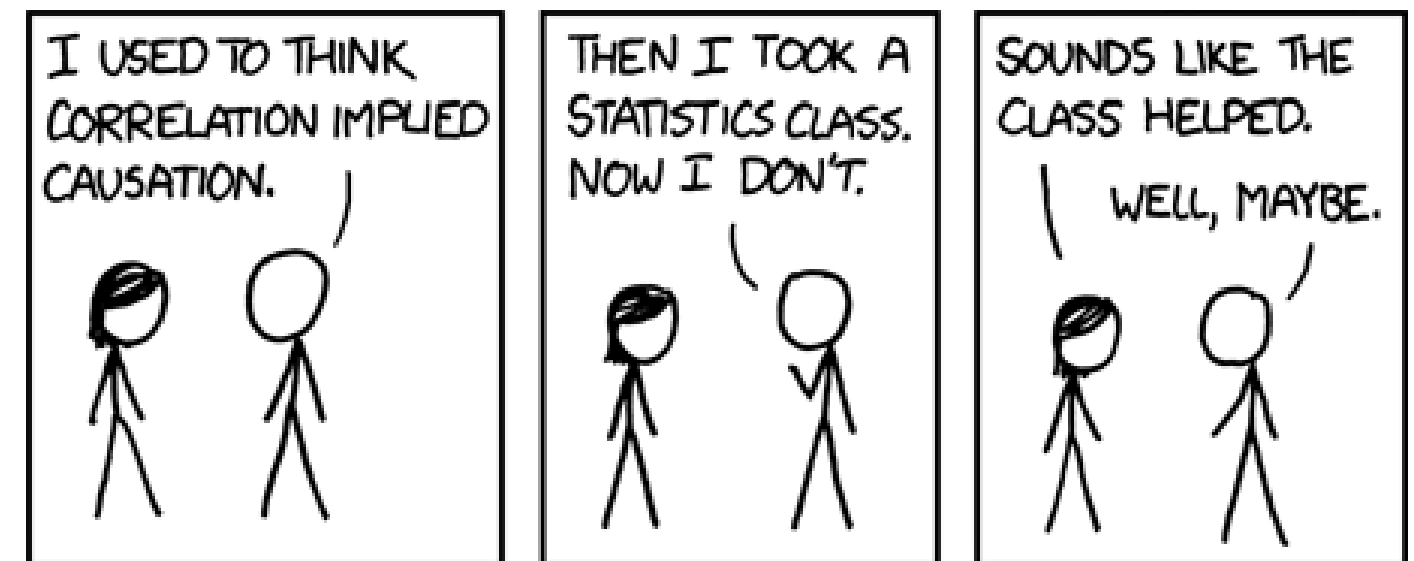
Association vs Causation

When two variables show some connection with one another, they are called **associated** variables.

- Associated variables can also be called **dependent** variables and vice versa.

If two variables are not associated, i.e., there is no evident connection between the two, then they are said to be **independent**.

In general, association does not imply causation, and causation can only be inferred from a randomized experiment.



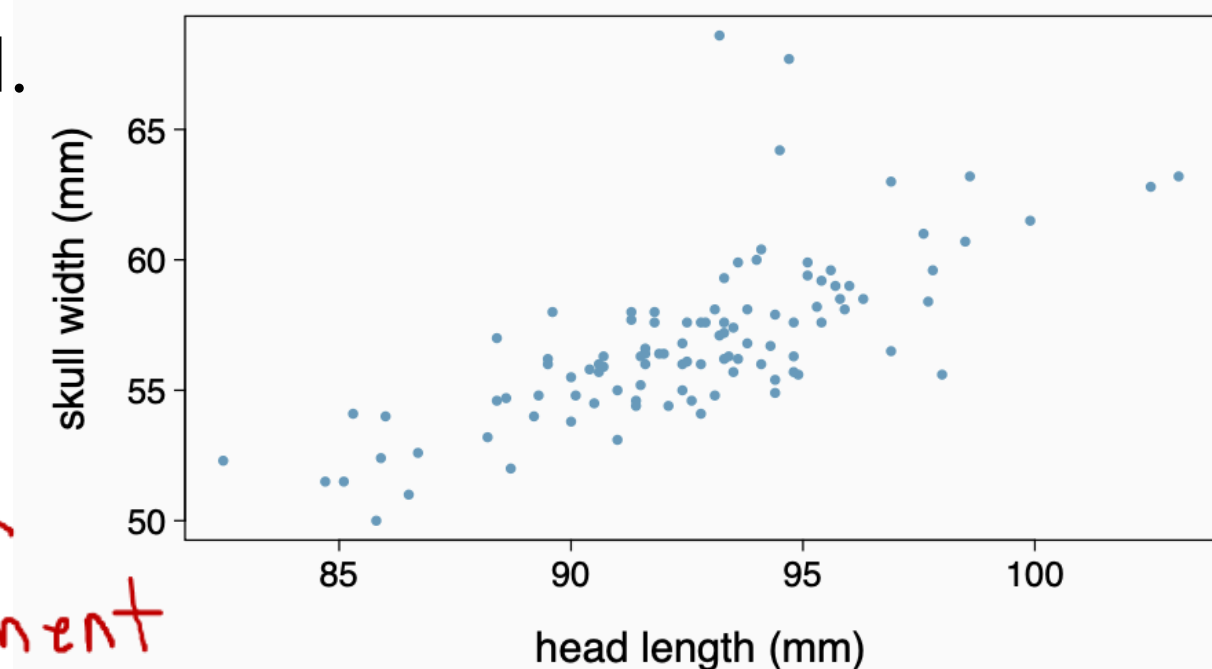
Association vs Causation

Let's try this problem:

Based on the scatterplot on the right, which of the following statements is correct about the head and skull lengths of possums?

- (a) There is no relationship between head length and skull width, i.e., the variables are independent.
- ☒ (b) Head length and skull width are positively associated.
- (c) Skull width and head length are negatively associated.
- (d) A longer head causes the skull to be wider.
- (e) A wider skull causes the head to be longer.

need further experiment



Association vs Causation

Let's try this problem (from Khan Academy):

longer name
taller

Slartibartfast, the king of Tralfamadore, was an exceptionally tall man. He had a hypothesis that there's a positive association in the land of Tralfamadore between the length of a person's first name and their height.

He ordered his servants to collect information about each and every person in the land, regarding their height and the length of their first name. Then, he analyzed the results and looked for trends in the data.

Q1: What type of statistical study did Slartibartfast use?

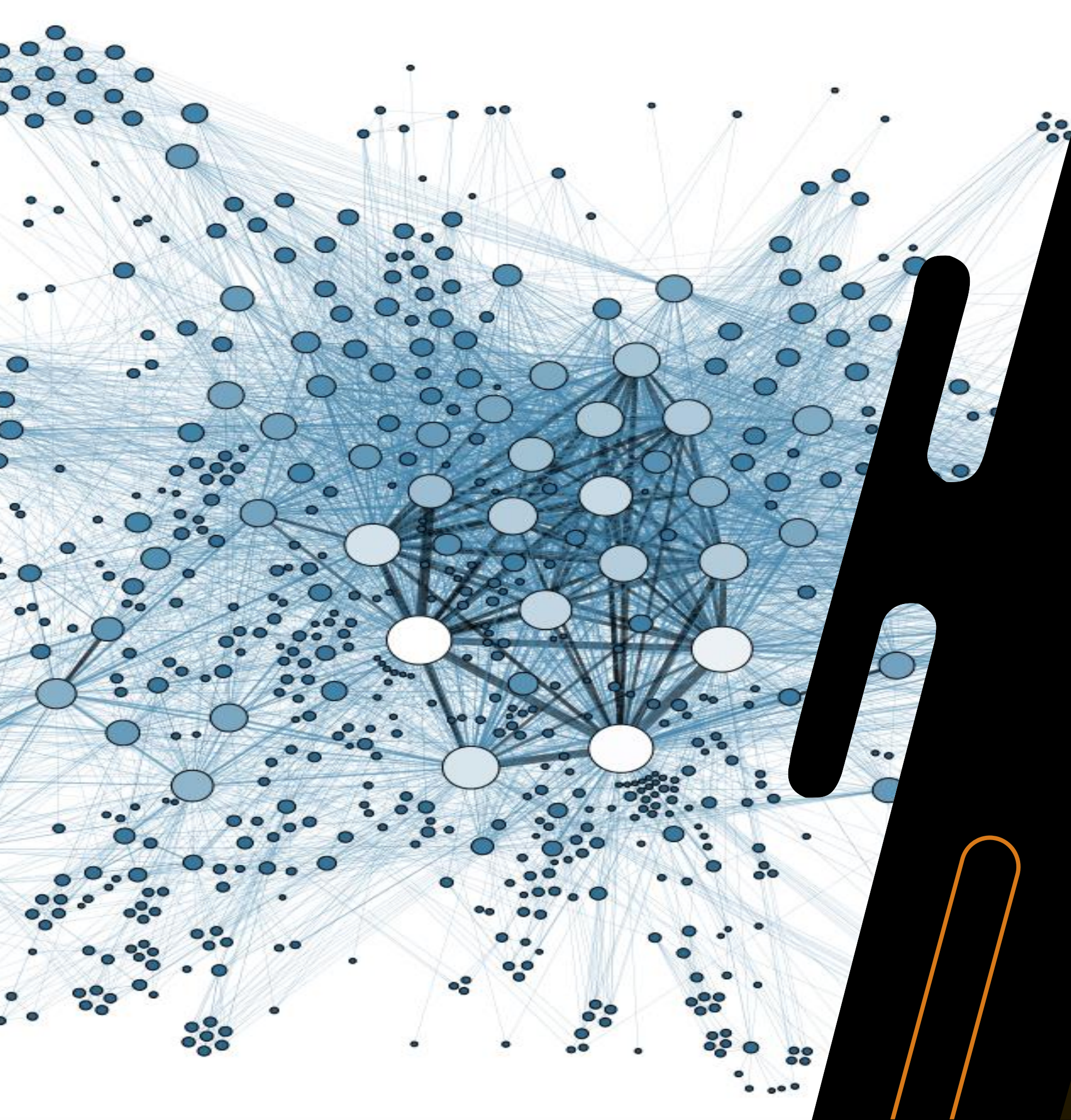
observational study

Q2: Is the study appropriate for the statistical questions it's supposed to answer?

Yes, he gathered the relevant (height, length of first name)

We will be back in
15 minutes at
7:15 CT





Sampling

Population vs Sample

Research question:

Can people make their bones stronger, merely by drinking milk?

Population of interest:

All people

Sample:

Group of adults who just started drinking milk regularly.



Population vs Sample

Each research question refers to a target population.

A sample: a subset of the cases referring to the question and is often a small fraction of the population.

We can generalize the population from the previous example:

Young adults, if the data are randomly sampled.

Anecdotal Evidence

If we consider this data collection situation:

Anti-smoking research started in the 1930s and 1940s when cigarette smoking became increasingly popular. While some smokers seemed to be sensitive to cigarette smoke, others were completely unaffected.

Anti-smoking research was faced with resistance based on **anecdotal evidence**, such as “*My uncle smokes three packs a day and he’s in perfectly good health*”.

It was concluded that “smoking is a complex human behavior, by its nature difficult to study, confounded by human variability.”

In time, researchers were able to examine larger samples of cases (smokers), and trends showing that smoking has negative health impacts became much clearer.

Anecdotal Evidence

Some other examples:

- I met two students who took more than 7 years to graduate from Duke, so it must take longer to graduate from Duke than from many other colleges.
- My friend's dad had a heart attack and died after they gave a new type of drug, so the drug must not work.

What problems do you see here?

1. The data only represent one or two cases.
2. It is unclear whether these cases are actually representative of the population.

>> Be careful of data collected in a haphazard fashion. They may be true and verifiable, but they may only represent extraordinary cases.

Anecdotal Evidence

Wouldn't it be better to just include everyone and “sample” the entire population?

Problems:

- It can be difficult to complete the whole population.
- Populations rarely stand still. They always change.
- It may be more complex than sampling.



Anecdotal Evidence

<https://www.pewresearch.org/short-reads/2020/07/24/how-removing-unauthorized-immigrants-from-census-statistics-could-affect-house-reapportionment/>

How removing unauthorized immigrants from census statistics could affect House reapportionment

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BY **JEFFREY S. PASSEL** AND **D'VERA COHN**



The House of Representatives. (National Archives)

Since the first census of the United States in 1790, counts that include both citizens and noncitizens have been used to [apportion seats](#) in the House of Representatives, with states gaining or losing based on population change over the previous decade. If unauthorized immigrants in the U.S. were removed from the 2020 census apportionment count – which the [White House seeks to do](#) – three states could each lose a seat they otherwise would have had and three others each could gain one, according to a Pew Research Center analysis based on government records.

Sampling from Population

Observational studies

Researchers collect data in a way that does not directly interfere with how the data arises.

Results of an observational study can generally be used to establish an association between the explanatory and response variables.



Sampling from Population

Why do we need good samples?

If observational data are not collected in a random framework from a population, these statistical methods – the estimates and errors associated with the estimates – are not reliable.

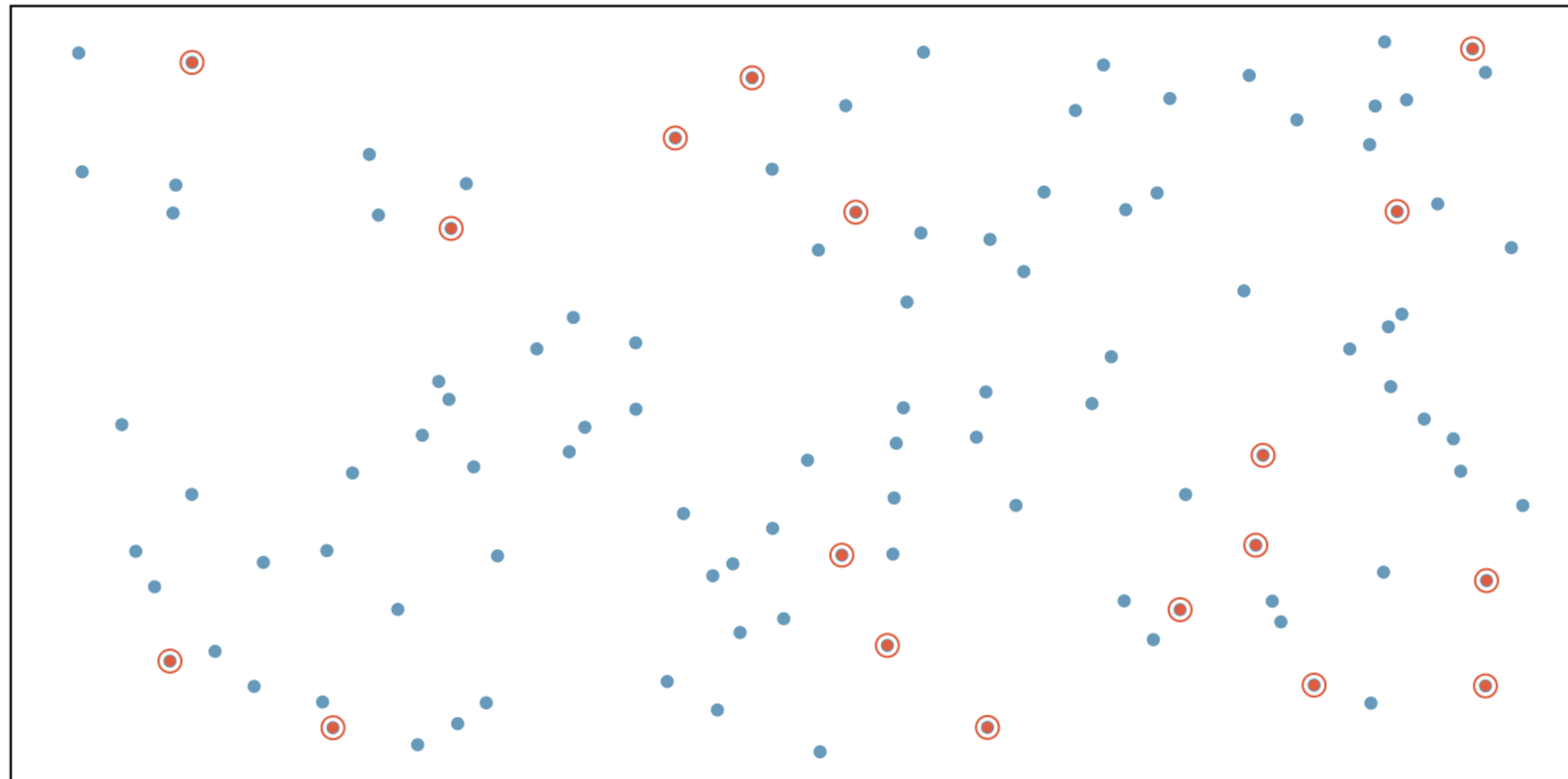
Four commonly used sampling techniques:

- Simple
- Stratified
- Cluster
- Multistage

Sampling from Population

Sampling Technique 1: Simple Random Sampling

Randomly select cases from the population, where there is no implied connection between the points that are selected.

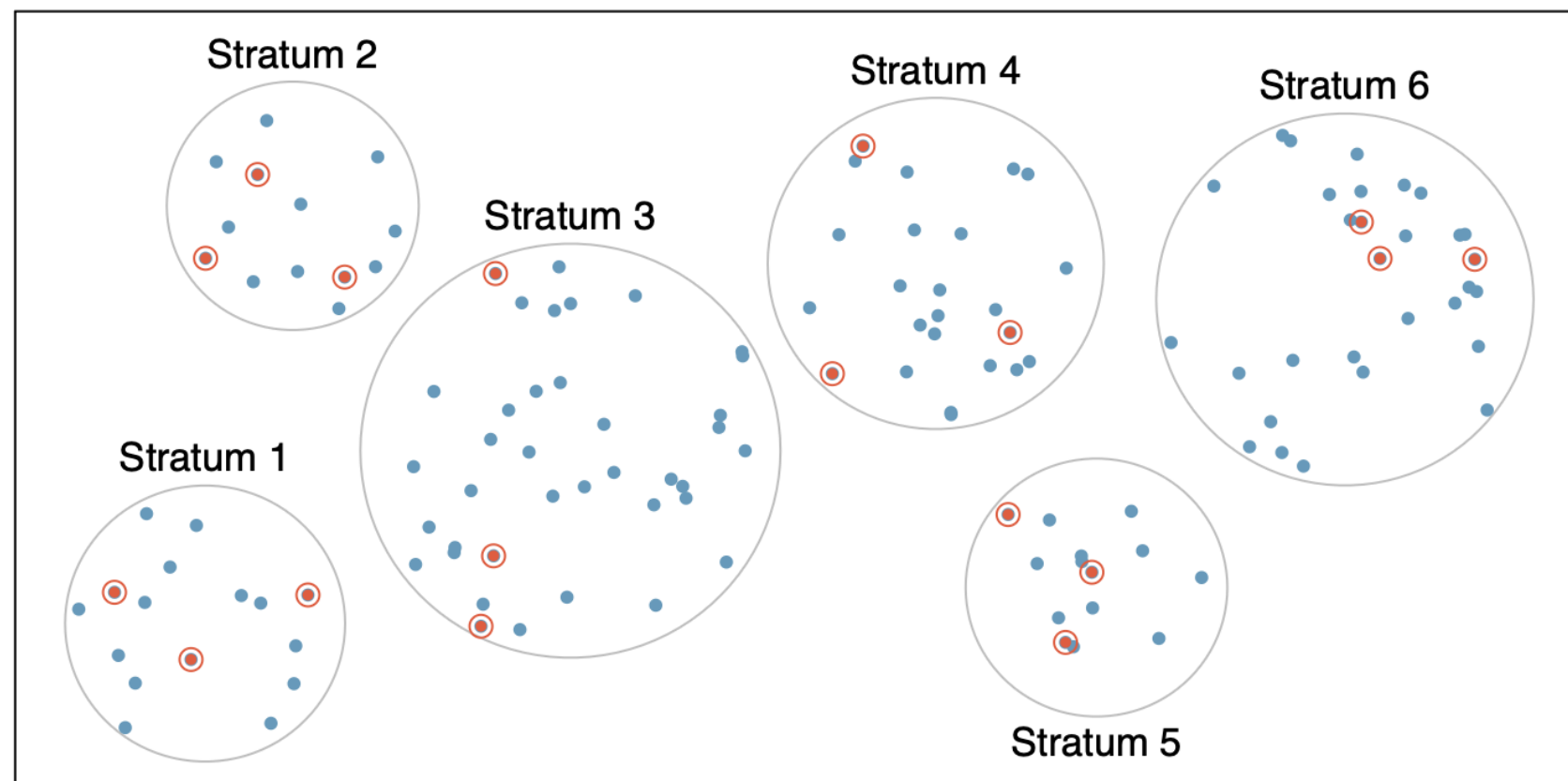


Sampling from Population

Sampling Technique 2: Stratified Sampling

A divide-and-conquer sampling strategy.

Strata are made up of similar observations. We take a simple random sample from each stratum.



Typically useful when the cases in each stratum are very similar with respect to the outcome of interest. But it might get more complex

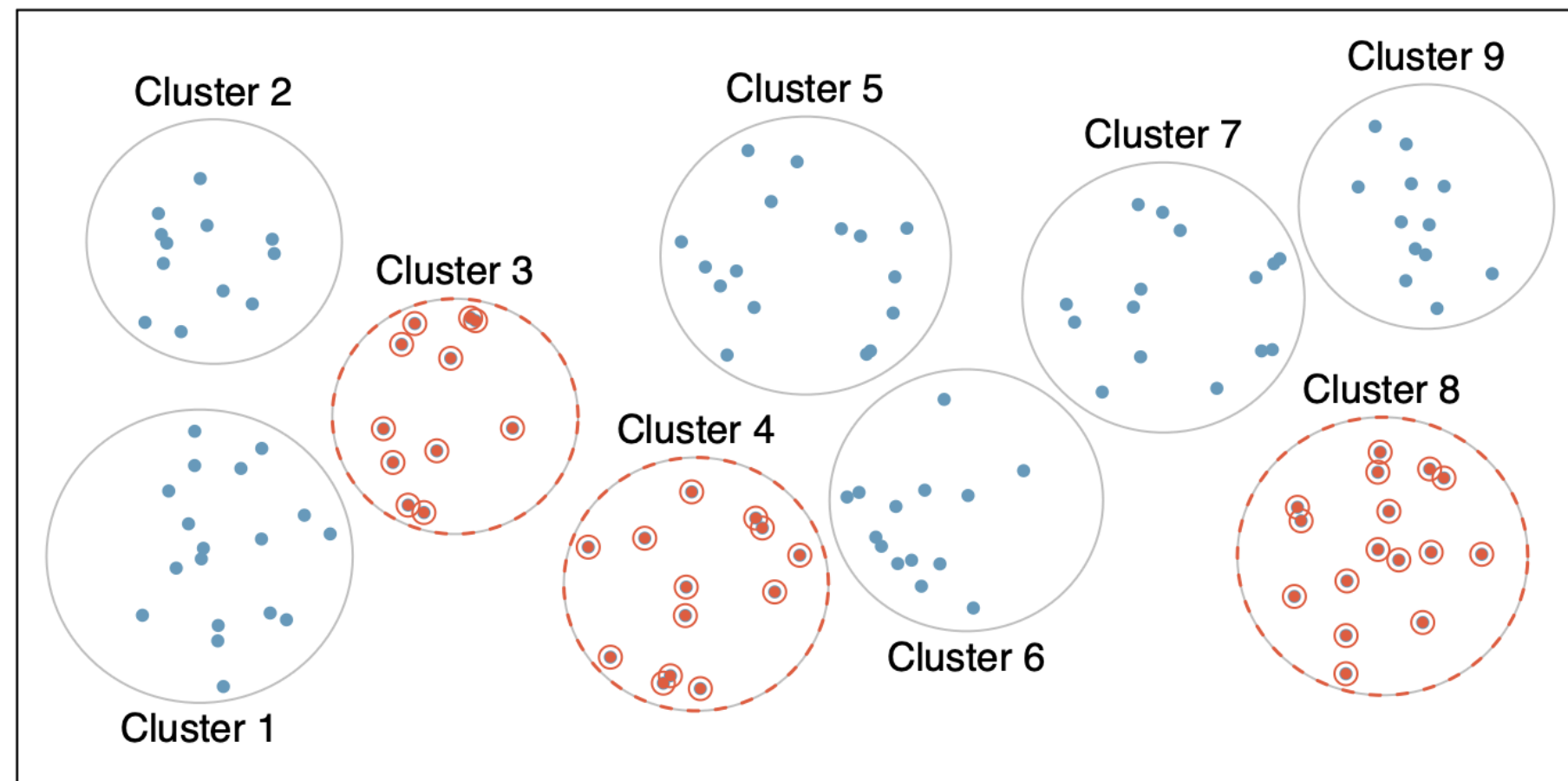
Sampling from Population

Sampling Technique 3: Cluster Sampling

The population is divided into many groups, called clusters.

Then, we sample a fixed number of clusters and include all observations from each of those clusters in the sample.

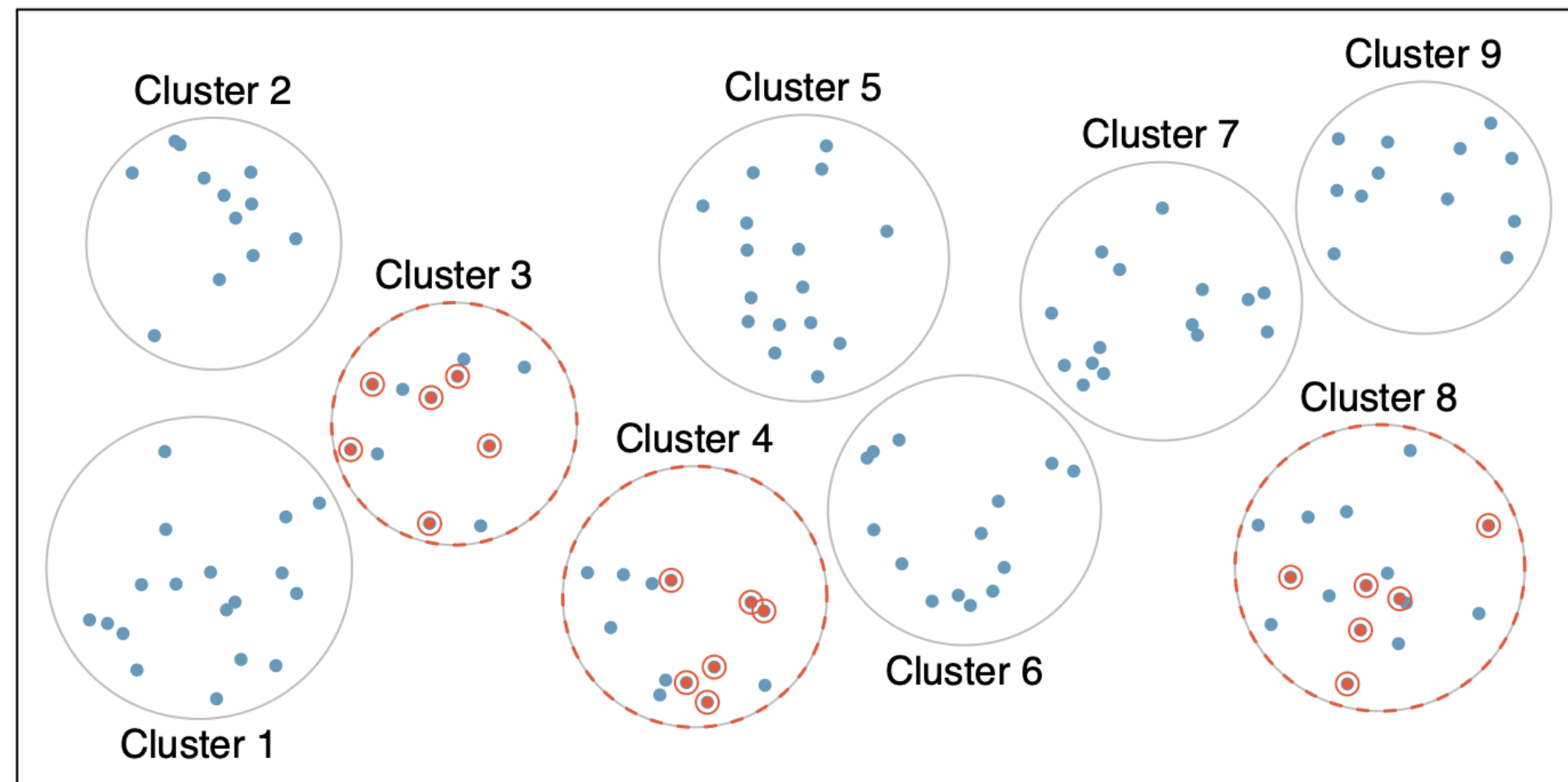
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Sampling from Population

Sampling Technique 4: Multistage Sampling

It is like a cluster sample, but rather than keeping all observations in each cluster, we collect a random sample within each selected cluster.



Sampling from Population

Which sampling method is better?

Cluster or multistage sampling techniques are usually preferred because

- more economical.
- more helpful when there is a lot of case-to-case variability within a cluster but the clusters themselves don't look very different from one another.

Downside: It may need more advanced techniques to analyze (or can be extensions of the methods that we are going to learn)

Sampling from Population

Let's try this problem:

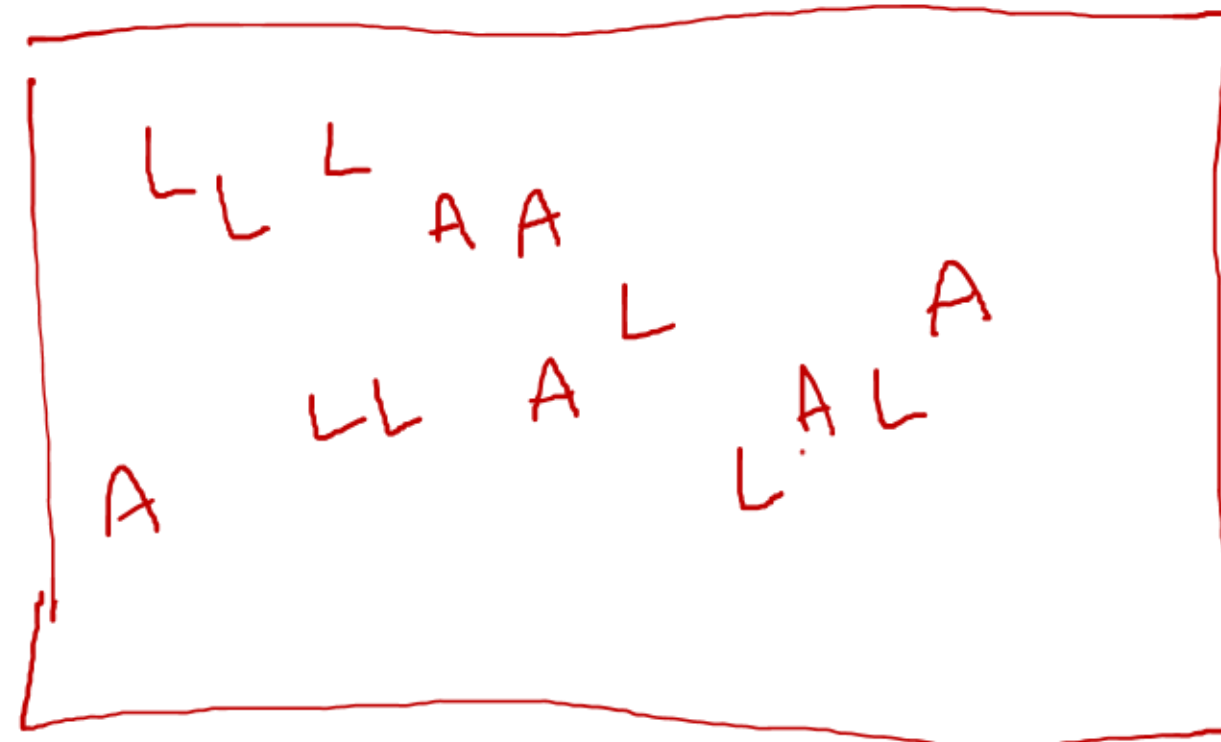
A city council has requested that a household survey be conducted in a suburban area of their city. The area is broken into many distinct and unique neighborhoods, some including large homes, some with only apartments. Which approach would likely be the least effective?

(a) Simple random sampling

(b) Cluster sampling

(c) Stratified sampling

(d) Blocked sampling



Sampling Bias



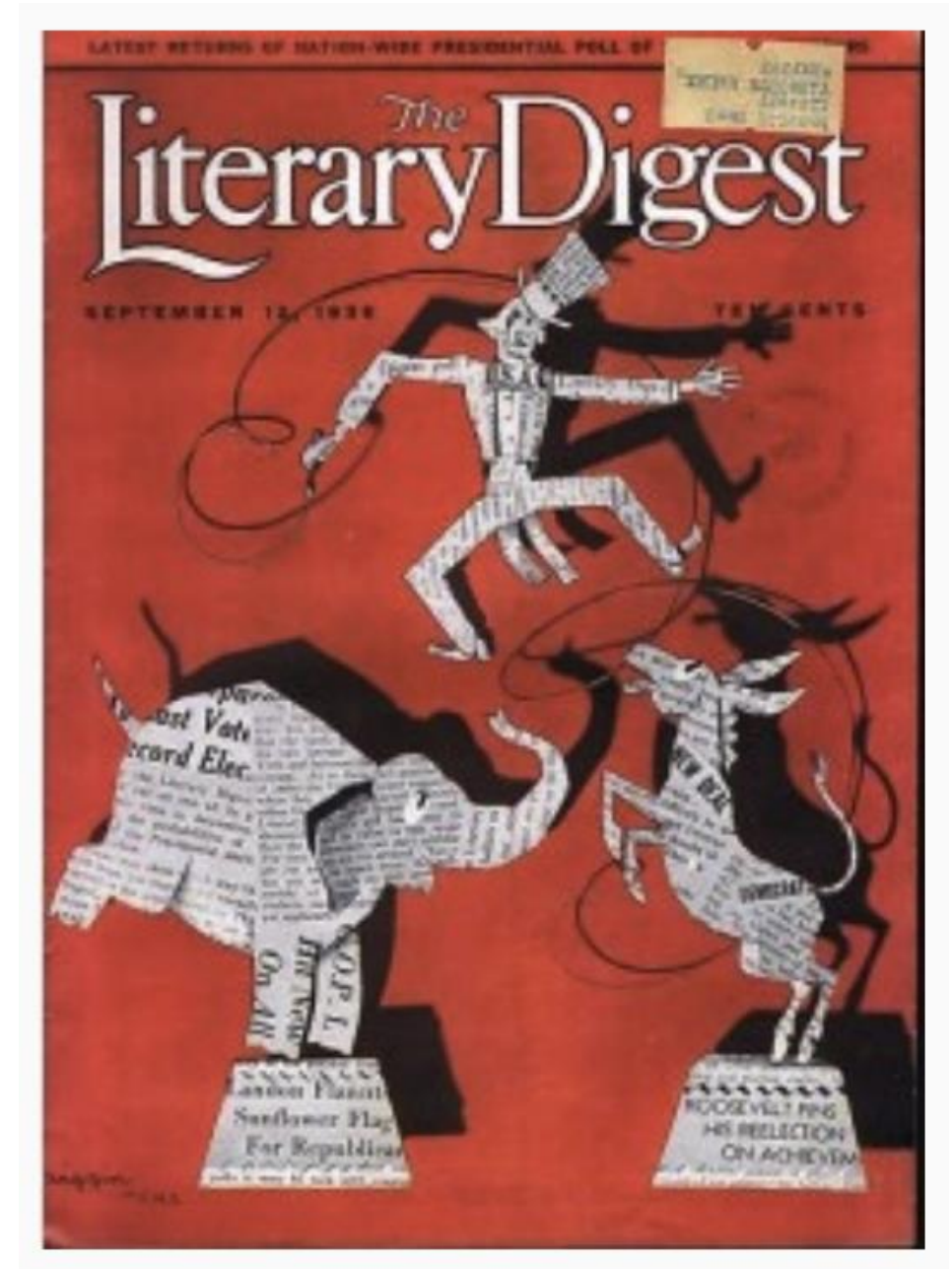
In 1936, Landon sought the Republican presidential nomination opposing the re-election of FDR.



Sampling Bias

The Literary Digest polled about 10 million Americans, and got responses from about 2.4 million.

- The poll showed that Landon would likely be the overwhelming winner and FDR would get only 43% of the votes.
- Election result: FDR won, with 62% of the votes.
- The magazine was completely discredited because of the poll, and was soon discontinued.



Sampling Bias

What went wrong?

The magazine had surveyed

- its own readers,
- registered automobile owners, and
- registered telephone users.

Large samples are preferable, but if the sample is biased, the sample will not yield an accurate prediction

These groups had incomes well above the national average of the day (remember, this is the Great Depression era), which resulted in lists of voters far more likely to support Republicans than a truly typical voter of the time, i.e., the sample was not representative of the American population at the time.

Sampling Bias

Large samples are preferable, but if the sample is biased, the sample will not yield an accurate prediction.

Another analogy:

If the soup is not well stirred, it doesn't matter how large a spoon you have, it will still not taste right. If the soup is well stirred, a small spoon will suffice to test the soup.

